

Supplemental Material for Resource-Certified Quantum-Huffman-Inspired Source Decoding with Group-Conditioned Measurements

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This Supplemental Material provides numerical protocols, certificate-feasibility diagnostics, extended tables, data-field definitions, and reproducibility details supporting the finite-instance results in the main text.

S1. Numerical and reproducibility overview

The numerical evidence has three roles. Numerical SDP calculations provide finite-instance primal/dual certificates where the grouped primal POVM is feasible and the fixed-length dual operator is feasible. The scalable PGM calculations are approximate diagnostics. The favorable-regime sweep is a finite sampled design-space map rather than a population-probability estimate. Source families are Zipf, Dirichlet, near-uniform, and strong-skew distributions. The benchmark uses alphabet sizes $M \in \{4, 8, 16, 32, 64\}$; the favorable sweep uses $M \in \{4, 8, 16, 32\}$. Payload modes are optimized, deterministic-phase, and random-phase. Grouping is random, probability-aware, or overlap-aware. The unweighted resource convention is $C_{\text{total}} = \bar{L} + C_A + C_{\text{QEC}}$, with declared alternatives reported in the associated CSV files.

S2. Data and file manifest

Scientific object	Data file	Generating script	Status
Default support-budget benchmark	data/benchmark_multiM.csv	run_benchmarks.py	PGM diagnostic
Payload comparison	data/payload_generation_comparison.csv	run_benchmarks.py	deterministic heuristic comparison
M=4 certificate sample	data/sdp_certificate_statistics.csv	run_certificate_search.py	numerical SDP certificate rows
Controlled M=8 probe	data/exact_sdp_m8_probe.csv	run_exact_sdp_m8_probe.py	numerical SDP certificate rows
Favorable-regime sweep	data/favorable_regime_sweep.csv	run_favorable_regime_sweep.py	PGM diagnostic
Sweep plots and summaries	data/favorable_regime_sweep_summary.csv	plot_favorable_regime_sweep.py	derived diagnostic summary
Support-overlap diagnostics	data/support_overlap_analysis.csv	support_overlap_analysis.py	derived diagnostic summary
Weighted sensitivity	data/resource_sensitivity.csv	run_resource_sensitivity.py	convention-sensitivity diagnostic
QEC hierarchy	data/qec_reliability_hierarchy.csv	run_qec_hierarchy.py	toy/proxy reliability diagnostic
Randomized validation	data/randomized_ensemble_validation.csv	experiments/run_randomized_ensemble_validation.py	SDP/PGM split by row
Helstrom validation	data/sdp_helstrom_validation.csv	experiments/validate_sdp_against_helstrom.py	numerical validation

The seed and configuration fields are retained in each source CSV. The full data dictionary and command mapping are provided in the accompanying reproducibility package.

S3. Support-budget and payload-generation protocol

For a positive source distribution, binary Huffman lengths are computed by a deterministic heap construction. The fixed-length reference cost is $n_{\text{fix}} = \lceil \log_2 M \rceil$, and the common dimension is $2^{n_{\text{max}}}$ with $n_{\text{max}} = \max(n_{\text{fix}}, \max_i l_i) +$

extra_qubits. Supports are cyclic computational-basis subsets with $|S_i| \leq 2^{n_{\max}-l_i}$. The offset is controlled by **support_overlap_shift**. Optimized states use the seeded local phase search in **payload_generation.py**; random-phase states use seeded phase draws, and deterministic-phase states use the deterministic phase construction. The optimized mode is a deterministic local-search heuristic, not a claim of global objective optimality.

S4. Grouping protocols

Random grouping uses a seeded random permutation and round-robin allocation. Probability-aware grouping assigns symbols in descending prior order to the group with lowest accumulated probability. Overlap-aware grouping assigns the next high-prior symbol to the group minimizing the probability-weighted within-group squared-overlap score plus the implemented 0.02 load penalty. The group count is declared per run; the benchmark uses $G = \min(\max(2, \lceil \sqrt{M} \rceil), M)$. Within a grouping comparison, the same payload ensemble and source probabilities are reused.

S5. SDP certificate construction and feasibility checks

For each numerical SDP, SCS solves the primal minimum-error POVM problem with tolerance **eps=5e-6** and at most 25000 iterations. The returned effects are Hermitized, projected to their positive semidefinite parts, regularized by $\max(10^{-10}, \mathbf{eps} \times 10^{-2})/m$, and normalized by the inverse square root of their sum. Their objective value is the feasible primal lower value $\underline{P}_{\text{group}}$. The primal completeness residual is $\|\sum_i E_i - I\|_F$, and the primal PSD residual is the magnitude of any negative minimum effect eigenvalue after postprocessing.

For the fixed-length reference, the dual variable is Hermitized and shifted by $\max(0, -\lambda_{\min}) + \max(10^{-10}, \mathbf{eps} \times 10^{-2})$ so that every $Y - p_i \rho_i$ is positive semidefinite. The trace of the shifted operator gives $\bar{P}_{\text{fix}}$. The dual feasibility residual is the magnitude of any negative minimum slack eigenvalue after the shift. The reported certificate gap is the sum of the grouped and fixed-side postprocessed primal-dual gaps. The implementation aborts if a postprocessed lower value exceeds its upper value by more than 10^{-8} . Thus the theorem is exact, whereas floating-point certificate validity relies on the reported numerical feasibility checks and postprocessing rather than symbolic or interval-arithmetic proof.

S6. Complete M=4 certificate table

The numerical-SDP certificate sample contains 40 $M = 4$ rows, of which three have $M_{\text{cert}} > 0$, giving $3/40 = 0.075$. The largest reported margin is 0.078835 after manuscript rounding. The separate five $M = 8$ PGM diagnostic rows in the same CSV have no numerical-SDP certificate values and are not included below.

Table 1: Complete $M = 4$ numerical-SDP certificate sample: instance metadata and accounting. Every row uses $C_A = 1$ and $n_{\text{fix}} = 2$; $C_{\text{Huff}} = \bar{L} + C_A$.

ID	Family	Parameter	Seed	Shift	$\bar{L}$	C_A	Grouping	Partition
M4-01	zipf	0.80	7480	0	1.890113	1	random	0,2 1,3
M4-02	zipf	0.80	7480	0	1.890113	1	overlap_aware	0 1,2,3
M4-03	zipf	0.80	7480	1	1.890113	1	random	0,2 1,3
M4-04	zipf	0.80	7480	1	1.890113	1	overlap_aware	0 1,2,3
M4-05	zipf	0.80	7511	0	1.890113	1	random	2,3 0,1
M4-06	zipf	0.80	7511	0	1.890113	1	overlap_aware	0 1,2,3
M4-07	zipf	0.80	7511	1	1.890113	1	random	2,3 0,1
M4-08	zipf	0.80	7511	1	1.890113	1	overlap_aware	0 1,2,3
M4-09	zipf	1.80	7580	0	1.483365	1	random	1,2 0,3
M4-10	zipf	1.80	7580	0	1.483365	1	overlap_aware	0 1,2,3
M4-11	zipf	1.80	7580	1	1.483365	1	random	1,2 0,3
M4-12	zipf	1.80	7580	1	1.483365	1	overlap_aware	0 1,2,3
M4-13	zipf	1.80	7611	0	1.483365	1	random	0,2 1,3
M4-14	zipf	1.80	7611	0	1.483365	1	overlap_aware	0 1,2,3
M4-15	zipf	1.80	7611	1	1.483365	1	random	0,2 1,3
M4-16	zipf	1.80	7611	1	1.483365	1	overlap_aware	0 1,2,3
M4-17	dirichlet	0.50	7450	0	1.754319	1	random	1,2 0,3
M4-18	dirichlet	0.50	7450	0	1.754319	1	overlap_aware	0,1 2,3
M4-19	dirichlet	0.50	7450	1	1.754319	1	random	1,2 0,3
M4-20	dirichlet	0.50	7450	1	1.754319	1	overlap_aware	0,1 2,3
M4-21	dirichlet	0.50	7481	0	1.330800	1	random	1,2 0,3
M4-22	dirichlet	0.50	7481	0	1.330800	1	overlap_aware	2 0,1,3
M4-23	dirichlet	0.50	7481	1	1.330800	1	random	1,2 0,3
M4-24	dirichlet	0.50	7481	1	1.330800	1	overlap_aware	2 0,1,3
M4-25	near_uniform	0.04	7404	0	2.000000	1	random	2,3 0,1
M4-26	near_uniform	0.04	7404	0	2.000000	1	overlap_aware	2,3 0,1
M4-27	near_uniform	0.04	7404	1	2.000000	1	random	2,3 0,1
M4-28	near_uniform	0.04	7404	1	2.000000	1	overlap_aware	2,3 0,1
M4-29	near_uniform	0.04	7435	0	2.000000	1	random	1,2 0,3
M4-30	near_uniform	0.04	7435	0	2.000000	1	overlap_aware	0,1 2,3
M4-31	near_uniform	0.04	7435	1	2.000000	1	random	1,2 0,3
M4-32	near_uniform	0.04	7435	1	2.000000	1	overlap_aware	0,1 2,3
M4-33	strong_skew	0.75	7475	0	1.416667	1	random	0,1 2,3
M4-34	strong_skew	0.75	7475	0	1.416667	1	overlap_aware	0 1,2,3
M4-35	strong_skew	0.75	7475	1	1.416667	1	random	0,1 2,3
M4-36	strong_skew	0.75	7475	1	1.416667	1	overlap_aware	0 1,2,3
M4-37	strong_skew	0.75	7506	0	1.416667	1	random	1,3 0,2
M4-38	strong_skew	0.75	7506	0	1.416667	1	overlap_aware	0 1,2,3
M4-39	strong_skew	0.75	7506	1	1.416667	1	random	1,3 0,2
M4-40	strong_skew	0.75	7506	1	1.416667	1	overlap_aware	0 1,2,3

Table 2: Complete $M = 4$ numerical-SDP certificate values and feasibility diagnostics. A dagger marks a positive finite-instance certificate. The residuals are defined in Sec. S5.

ID	$\underline{P}_{\text{group}}$	$\bar{P}_{\text{fix}}$	Group lower/cost	Fixed upper/cost	M_{cert}	Primal residual	Dual residual	Gap
M4-01	0.966541	0.865707	0.334430	0.432853	-0.098423	5.15e-15	0.00e+00	3.23e-05
M4-02	0.939454	0.865707	0.325058	0.432853	-0.107796	3.03e-15	0.00e+00	2.25e-05
M4-03	1.000000	0.979742	0.346007	0.489871	-0.143864	3.61e-15	0.00e+00	3.14e-05
M4-04	0.999997	0.979742	0.346006	0.489871	-0.143865	3.61e-15	0.00e+00	2.64e-05
M4-05	0.999996	0.891779	0.346006	0.445890	-0.099884	3.00e-15	0.00e+00	5.81e-05
M4-06	0.939454	0.891779	0.325058	0.445890	-0.120832	4.05e-15	0.00e+00	2.60e-05
M4-07	0.979731	0.979742	0.338994	0.489871	-0.150877	3.50e-15	0.00e+00	3.40e-05
M4-08	0.999997	0.979742	0.346006	0.489871	-0.143865	3.50e-15	0.00e+00	2.64e-05

ID	P_{group}	$\bar{P}_{\text{fix}}$	Group lower/cost	Fixed upper/cost	M_{cert}	Primal residual	Dual residual	Gap
M4-09	0.964589	0.938450	0.388420	0.469225	-0.080804	3.15e-15	0.00e+00	2.69e-05
M4-10	0.964591	0.938450	0.388421	0.469225	-0.080804	3.15e-15	0.00e+00	2.13e-05
M4-11	0.999999	0.981092	0.402679	0.490546	-0.087867	3.46e-15	0.00e+00	8.81e-06
M4-12	1.000000	0.981092	0.402679	0.490546	-0.087867	3.46e-15	0.00e+00	8.19e-06
M4-13	0.979274	0.938449	0.394333	0.469225	-0.074891	4.83e-15	0.00e+00	2.33e-05
M4-14	0.964591	0.938449	0.388421	0.469225	-0.080804	4.83e-15	0.00e+00	2.11e-05
M4-15	1.000000	0.981092	0.402679	0.490546	-0.087867	3.61e-15	0.00e+00	2.34e-05
M4-16	1.000000	0.981092	0.402679	0.490546	-0.087867	4.50e-15	0.00e+00	8.20e-06
M4-17	0.957081	0.862966	0.347483	0.431483	-0.084000	2.81e-15	0.00e+00	5.02e-06
M4-18	0.925615	0.862966	0.336059	0.431483	-0.095424	3.60e-15	0.00e+00	5.61e-06
M4-19	1.000000	0.977695	0.363066	0.488848	-0.125782	4.53e-15	0.00e+00	1.62e-06
M4-20	1.000000	0.977695	0.363066	0.488848	-0.125782	4.53e-15	0.00e+00	8.69e-06
M4-21	0.972206	0.952700	0.417113	0.476350	-0.059238	3.58e-15	0.00e+00	4.15e-06
M4-22	0.972207	0.952700	0.417113	0.476350	-0.059237	3.27e-15	0.00e+00	1.75e-06
M4-23	1.000000	0.982121	0.429037	0.491060	-0.062023	4.08e-15	0.00e+00	1.41e-05
M4-24	1.000000	0.982121	0.429037	0.491060	-0.062023	2.84e-15	0.00e+00	1.27e-05
M4-25	1.000000	1.000005	0.333333	0.500002	-0.166669	5.77e-16	0.00e+00	1.84e-05
M4-26	1.000000	1.000005	0.333333	0.500002	-0.166669	5.77e-16	0.00e+00	1.84e-05
M4-27 [†]	1.000000	0.508997	0.333333	0.254498	0.078835	4.00e-16	0.00e+00	1.83e-05
M4-28 [†]	1.000000	0.508997	0.333333	0.254498	0.078835	4.00e-16	0.00e+00	1.83e-05
M4-29	1.000000	1.000009	0.333333	0.500005	-0.166671	4.58e-16	0.00e+00	3.40e-05
M4-30	1.000000	1.000009	0.333333	0.500005	-0.166671	5.21e-16	0.00e+00	1.69e-05
M4-31	0.516886	0.516890	0.172295	0.258445	-0.086150	3.14e-16	0.00e+00	4.83e-06
M4-32 [†]	1.000000	0.516890	0.333333	0.258445	0.074888	5.66e-16	0.00e+00	8.24e-06
M4-33	0.980807	0.958018	0.405851	0.479009	-0.073158	3.99e-15	0.00e+00	4.14e-05
M4-34	1.000000	0.958018	0.413793	0.479009	-0.065216	2.91e-15	0.00e+00	3.39e-05
M4-35	0.980807	0.969729	0.405851	0.484864	-0.079013	3.04e-15	0.00e+00	4.02e-05
M4-36	1.000000	0.969729	0.413793	0.484864	-0.071071	2.83e-15	0.00e+00	3.27e-05
M4-37	0.980807	0.958018	0.405851	0.479009	-0.073158	2.79e-15	0.00e+00	4.14e-05
M4-38	1.000000	0.958018	0.413793	0.479009	-0.065216	3.14e-15	0.00e+00	3.39e-05
M4-39	1.000000	0.969729	0.413793	0.484864	-0.071071	3.73e-15	0.00e+00	3.35e-05
M4-40	1.000000	0.969729	0.413793	0.484864	-0.071071	3.73e-15	0.00e+00	3.27e-05

S7. Controlled M=8 certificate probe

The two controlled numerical-SDP instances yield $M_{\text{cert}} = 0.081398$ and $M_{\text{cert}} = -0.083336$, respectively. They are finite-instance probes and do not constitute a broad $M = 8$ certificate survey.

Table 3: Controlled $M = 8$ numerical-SDP probe. Both rows have $\bar{L} = n_{\text{fix}} = 3$ and $C_A = 1$.

Seed	Payload/grouping	Shift	M_{cert}	$\underline{P}_{\text{group}}$	$\bar{P}_{\text{fix}}$	Primal res.	Dual res.	Gap	Runtime (s)
881	optimized/overlap_aware	1	0.081398	0.999998	0.505806	7.77e-16	0.00e+00	8.75e-06	0.197
993	deterministic_phase/random	0	-0.083336	0.999999	1.000006	1.01e-15	0.00e+00	3.47e-05	0.119

S8. Favorable-regime sweep

The favorable sweep samples $M \in \{4, 8, 16, 32\}$, the source families and parameter grids encoded in `run_favorable_regime_sweep.py`, overlap shifts 0, 1, 2, 4, payload modes optimized/deterministic-phase/random-phase, and the three grouping protocols. The QEC labels are `no_qec`, `low_overhead_proxy`, and `moderate_overhead_proxy`, with $C_{\text{QEC}} = 0, 1, 3$. It uses PGM, and every row is labeled as a diagnostic. The complete 56160-row CSV remains in the reproducibility package. For the no-QEC subset, the following means

Table 4: Verified favorable-regime sweep counts. Fractions are empirical summaries of the finite sampled design space.

Analysis	Rows	Positive rows	Fraction
All QEC-inclusive rows	56160	6828	0.122
No-QEC PGM subset	18720	4812	0.257
Weighted sweep rows	56160	19472	0.347

summarize the positive and nonpositive diagnostic rows.

Class	G_{comp}	g_{rel}	r_{sep}	r_{cost}	R_{sup}	A_{Huff}
Positive	2.5221	0.5771	0.8339	0.3564	0.3732	0.001659
Nonpositive	0.9288	0.2557	0.7966	0.9783	0.2442	0.005424

These are finite-sample diagnostic summaries.

S9. Support-overlap diagnostics

The support-overlap ratio R_{sup} is computed from intersections of assigned computational-basis supports. It is distinct from the payload-state Gram overlap and from minimum-error success. The ambiguity quantities $A_{\text{Huff}} = A_{\text{global}}$, A_{in} , and $A_{\text{sep}} = A_{\text{global}} - A_{\text{in}}$ are computed from source-weighted squared payload overlaps. The ratio $r_{\text{sep}} = A_{\text{sep}}/A_{\text{global}}$ is recorded with a zero-denominator guard. The derived statistics are stored in `data/support_overlap_analysis.csv` and `data/support_overlap_summary.csv`.

S10. Weighted resource sensitivity

The default sensitivity scan varies $w_A/w_L \in \{0, 0.5, 1, 1.5, 2, 3\}$, $w_Q/w_L \in \{0, 0.25, 0.5, 1, 1.5, 2\}$, and $w_D/w_L \in \{0, 0.01, 0.03, 0.06, 0.1\}$, while setting $w_M/w_L = 0.01$ and $w_G/w_L = 0.002$. A weighted row is positive when $M_{\text{res}}^{(w)} > 0$. The default scan has 181 positive rows out of 9540, or 0.01897 (1.90%). The favorable sweep’s weighted-positive fraction is 0.34672 (0.347). This is a declared convention-sensitivity analysis, not a hardware-calibrated theorem.

S11. QEC reliability proxy details

The QEC layer uses only the implemented repetition-style toy model, stabilizer-code proxy, and depth-sensitive proxy. Under the stated independent-error family assumptions, $P_{\text{corr}} \geq \sum_{w=0}^t \binom{n}{w} p^w (1-p)^{n-w}$ and $q_L = 1 - P_{\text{corr}}$, hence $q_L \leq 1 - \sum_{w=0}^t \binom{n}{w} p^w (1-p)^{n-w}$. The reported C_{QEC} values are declared resource proxies. These calculations are not hardware-calibrated fault-tolerant simulations.

S12. Randomized ensemble validation

The randomized validation script uses a fixed generator seed, Dirichlet source draws, normalized complex-Gaussian pure states in dimension four, and alphabet sizes $M \in \{4, 8, 16, 32\}$. It evaluates random, probability-aware, and overlap-aware partitions. Small cases may use numerical SDP according to the script's solver path, while larger rows use PGM. The CSV records the relevant baseline type and row-level configuration.

S13. Figure and table provenance

Item	Output	Data source	Script or construction
Fig. 1	TikZ pipeline	none	main.tex
Figs. 2–5	support saving, payload, efficiency, grouping	benchmark and payload CSVs	make_figures.py
Figs. 6–9	favorable-sweep diagnostics	favorable sweep CSV	plot_favorable_regime_sweep.py
Figs. 10–11	M=4 certificate distribution and fraction	certificate statistics CSV	make_figures.py
Fig. 12	controlled M=8 probe	M=8 probe CSV	run_exact_sdp_m8_probe.py
Fig. 13	weighted sensitivity	resource sensitivity CSV	make_figures.py
Figs. E1–E2	additional phase diagrams	enhanced phase CSV	experiments/run_appendix_phase_diagrams.py
Figs. G1–G3	overlap, Zipf, support-overlap	grouping, Zipf, overlap CSVs	grouping, Zipf, and support-overlap scripts
Figs. G4–G5	reliability and communication	QEC and communication CSVs	QEC hierarchy and communication scripts
Figs. G6–G7	randomized diagnostics	randomized validation CSV	experiments/run_randomized_ensemble_validation.py
Figs. G8–G10	support, ambiguity, certificate diagnostics	four-symbol and certificate CSVs	four-symbol and certificate scripts
Figs. G11–G12	representative M=4 certificates	certificate statistics CSV	make_figures.py

S14. Reproduction commands

The reproducibility package provides the full commands and expected output paths. In brief, install the packages in `code/requirements.txt`, run the selected scripts from the package's `code/` directory, and build the manuscripts with `latexmk -pdf main.tex` and `latexmk -pdf supplemental.tex`. The principal entries are `run_benchmarks.py`, `run_certificate_search.py`, `run_exact_sdp_m8_probe.py`, `run_favorable_regime_sweep.py`, `run_resource_sensitivity.py`, `run_qec_hierarchy.py`, `make_figures.py`, and `plot_favorable_regime_sweep.py`.